Unlock the Lab: sample studies

Six studies for discussion, with an answer key for the facilitator

Pablo Bernabeu

About this set

These six fictional studies come from Unlock the Lab and are printed as participants see them in the app, with the two answers that the app asks for. Use them for discussion in groups, as a warm-up before the app or for anyone taking part on paper. The evaluation guide has the rubric, the advice on reaching one rating and the glossary to go with them.

The set covers the three quality bands of the expert ratings: Low quality (expert rating 1–2), Medium quality (expert rating 3–5) and High quality (expert rating 6–7). From each band it takes the study that is hardest to judge, with the most criteria whose band differs from the overall one, and the clearest, with the fewest, where possible with a different expert rating. The studies keep their order in the app, so their letters say nothing about their quality.

Each study is on its own page. The answer key at the end gives the expert's rating, a summary and a note on each criterion; keep it back until everyone has answered. To play 'Predict the crowd' on paper, collect the ratings, work out the group's average for each study and compare it with the predictions. The live dashboard shows how participants have rated the current version of each study in the app, under the dashboard number given in the answer key.

Study A · Fictional study

TITLE

Shocking study proves chocolate for breakfast turns children into maths stars

Access

Free to everyone online: anyone can read the full paper, with no sign-up.

Source

The journal charged the authors £1,800 and published the paper five days after it arrived. Its website names no editors and says nothing about peer review.

Theory

The study asks whether chocolate at breakfast raises children's maths scores. It treats sugar as instant fuel for the brain. It does not mention earlier research on the 'sugar crash' (tiredness after a burst of sugar).

Methods & Data

On three mornings, *N*=12 children in one classroom ate chocolate for breakfast and then took a maths test. There was no control group eating an ordinary breakfast. **Data availability:** The test scores are not shared, for 'commercial' reasons linked to the chocolate brand, the authors say.

Conclusion

The authors conclude that chocolate breakfasts make children better at maths. They call for chocolate at every school breakfast to raise national exam results.

Your answers

1. Predict the crowd

What will the average rating from all workshop participants be? Your score compares your prediction with a target that also counts the expert's rating heavily.

Very low (1) (2) (3) (4) (5) (6) (7) Very high

2. Your own rating

How good is this study overall? Weigh up the six criteria.

Very low (1) (2) (3) (4) (5) (6) (7) Very high

Study B · Fictional study

TITLE

Field trial finds no clear effect of one pesticide on bees' ability to fly home, at current farm doses

Access

Free to everyone online: anyone can download the full paper.

Source

It appeared in a peer-reviewed ecology journal run by a large scientific society. The pesticide's maker paid for half the trial and saw the results before publication. The authors declare this conflict of interest.

Theory

Lab studies found that this pesticide made bees lose their way. This study asks whether that happens on real farms at the doses farmers use.

Methods & Data

Over two years, *N*=6,000 bees on 20 farms wore small radio tags recording whether they flew home. Half the farms, picked at random, used the pesticide, following a preregistered plan. **Data availability:** All data and code are free online.

Conclusion

The authors report no clear harm to bees' ability to fly home at current farm doses. They say a small effect is possible and warn against applying the result to other insects.

Your answers

1. Predict the crowd

What will the average rating from all workshop participants be? Your score compares your prediction with a target that also counts the expert's rating heavily.

Very low (1) (2) (3) (4) (5) (6) (7) Very high

2. Your own rating

How good is this study overall? Weigh up the six criteria.

Very low (1) (2) (3) (4) (5) (6) (7) Very high

Study C · Fictional study

TITLE

Instagram use raises anxiety in teenagers, survey finds

Access

Not free to everyone: the full paper needs a university library login.

Source

It appeared in a peer-reviewed psychology journal. The journal's website says it wants 'findings that get people talking'. It sends press releases to journalists for most papers.

Theory

The study uses the idea of social comparison: people can feel worse after comparing themselves with others. It assumes Instagram use leads directly to anxiety and does not discuss other causes of teenage anxiety.

Methods & Data

The researchers surveyed *N*=5,000 teenagers once (a cross-sectional study). Teenagers reported their own hours on Instagram and their own anxiety. **Data availability:** Summary figures are in the paper; raw data are available from the lead author on request.

Conclusion

The authors report that teenagers who used Instagram more reported more anxiety. They mostly call this a link but hint in places that Instagram is the cause. They do not ask whether anxious teenagers simply use social media more.

Your answers

1. Predict the crowd

What will the average rating from all workshop participants be? Your score compares your prediction with a target that also counts the expert's rating heavily.

Very low (1) (2) (3) (4) (5) (6) (7) Very high

2. Your own rating

How good is this study overall? Weigh up the six criteria.

Very low (1) (2) (3) (4) (5) (6) (7) Very high

Study D · Fictional study

TITLE

Male bosses of the 500 largest US companies are taller than the average US man

Access

Free to everyone online as a preprint (an early paper shared before review).

Source

Posted on a public preprint server. It has not been peer reviewed or accepted by a journal.

Theory

The authors argue that early hunter-gatherers chose tall men as leaders. They say the same instinct still picks company bosses but give no evidence. The paper does not mention the ‘halo effect’ (when one liked trait, such as height, colours how we judge someone).

Methods & Data

The authors collected the heights of *N*=452 men who led the 500 largest US companies. They compared these with the national average but did not measure profits or leadership skill. **Data availability:** A spreadsheet of heights is available from the authors on request.

Conclusion

The bosses were about 6 cm taller than average. The authors conclude that tall men are born to lead and that evolution explains why companies choose them.

Your answers

1. Predict the crowd

What will the average rating from all workshop participants be? Your score compares your prediction with a target that also counts the expert’s rating heavily.

Very low 1 2 3 4 5 6 7 Very high

2. Your own rating

How good is this study overall? Weigh up the six criteria.

Very low 1 2 3 4 5 6 7 Very high

Study E · Fictional study

TITLE

Systematic review of 38 trials: mindfulness brings small to moderate chronic pain relief

Access

Free to everyone online: anyone can read the full review and download its data.

Source

Published in a library of health reviews run by an international non-profit research network. Each review's plan and final text are peer reviewed.

Theory

Many randomised controlled trials have tested mindfulness (training attention on the present moment) for chronic (long-lasting) pain, with mixed results. This review combines them, drawing on research into how the brain handles pain.

Methods & Data

Following a preregistered plan, the team found 38 suitable trials with *N*=3,540 participants and combined them in a meta-analysis. Two reviewers working separately checked each trial for bias. **Data availability:** All extracted data and code are free online.

Conclusion

The authors report a small to moderate drop in pain (effect size 0.33) that varied widely between trials. Mindfulness helped some kinds of pain more than others. They say it can support medical treatment but not replace it.

Your answers

1. Predict the crowd

What will the average rating from all workshop participants be? Your score compares your prediction with a target that also counts the expert's rating heavily.

Very low (1) (2) (3) (4) (5) (6) (7) Very high

2. Your own rating

How good is this study overall? Weigh up the six criteria.

Very low (1) (2) (3) (4) (5) (6) (7) Very high

Study F · Fictional study

TITLE

Real-world study of 5,000 people finds PrEP stops HIV infection

Access

Free to everyone online: anyone can read the paper.

Source

Published in a peer-reviewed infectious disease journal run by a large medical society. A government health agency funded the study.

Theory

PrEP, most often a daily pill, prevents infection with HIV (the virus that causes AIDS). Earlier randomised controlled trials found it works when taken as prescribed. This study asks how well it works in everyday life.

Methods & Data

This cohort study followed *N*=5,000 people at high risk of HIV, using blood tests to check they took PrEP. They were compared with similar patients in past clinic records (historical controls), not a group chosen by chance. **Data availability:** On request from the authors.

Conclusion

The authors report almost complete protection with daily use, falling sharply with missed doses. They note PrEP does not prevent other sexually transmitted infections. Only an extra online file mentions that the past patients may have differed in other ways.

Your answers

1. Predict the crowd

What will the average rating from all workshop participants be? Your score compares your prediction with a target that also counts the expert's rating heavily.

Very low (1) (2) (3) (4) (5) (6) (7) Very high

2. Your own rating

How good is this study overall? Weigh up the six criteria.

Very low (1) (2) (3) (4) (5) (6) (7) Very high

For the facilitator

Answer key

The expert's feedback is the same as the app shows after each study.

Study	Title	Expert rating	Quality	Dashboard number
Study A	Shocking study proves chocolate for breakfast turns children into maths stars	1	Low quality	1
Study B	Field trial finds no clear effect of one pesticide on bees' ability to fly home, at current farm doses	5	Medium quality	5
Study C	Instagram use raises anxiety in teenagers, survey finds	4	Medium quality	6
Study D	Male bosses of the 500 largest US companies are taller than the average US man	2	Low quality	11
Study E	Systematic review of 38 trials: mindfulness brings small to moderate chronic pain relief	7	High quality	24
Study F	Real-world study of 5,000 people finds PrEP stops HIV infection	6	High quality	41

Answer key

Study A: Shocking study proves chocolate for breakfast turns children into maths stars

Expert rating: 1 out of 7 · Low quality

Five of the six criteria are weak. The journal took a fee, published within five days and names no editors. Twelve children ate chocolate with no control group, the scores are hidden and the authors still call for chocolate at every school breakfast. Free access cannot make up for this, so it sits at the bottom of the scale.

Criterion by criterion

Criterion	Band	Note
Title	Low	‘Shocking’, ‘proves’ and ‘maths stars’ are press-release hype and promise far more than a test of 12 children on three mornings could show.
Access	High	Anyone can read the full paper free online with no sign-up; Access is about how open a paper is, not whether the study is any good.
Source	Low	A £1,800 fee, publication five days after the paper arrived, no named editors and no word on peer review point to a predatory journal.
Theory	Low	It treats sugar as instant fuel for the brain and ignores earlier research on the ‘sugar crash’, so it does not build on what is already known.
Methods & Data	Low	Only 12 children in one class were tested on three mornings, with no control group on an ordinary breakfast and no shared scores.
Conclusion	Low	Without a comparison group the test cannot show that chocolate caused anything, yet the authors claim it does and want it at every school breakfast.

Answer key

Study B: Field trial finds no clear effect of one pesticide on bees' ability to fly home, at current farm doses

Expert rating: 5 out of 7 · Medium quality

Free access, an exact title, a randomised and preregistered field trial with open data and a careful conclusion stand out. It is held back because the pesticide's maker paid for half the trial and saw the results before publication, so readers cannot rule out that this shaped what was reported, even though the authors declared it.

Criterion by criterion

Criterion	Band	Note
Title	High	It names the design, the single pesticide, the finding and its key limit, 'at current farm doses', without overstating anything.
Access	High	Anyone can download the full paper free online, so every reader can check the methods for themselves.
Source	Medium	The society journal uses peer review, but the pesticide's maker paid for half the trial and saw the results first, a declared conflict of interest.
Theory	High	It fills a clear gap by testing whether the lab finding that the pesticide made bees lose their way holds up on real farms.
Methods & Data	High	The farms using the pesticide were picked at random, the plan was preregistered, 6,000 bees were tracked over two years and all data and code are free online.
Conclusion	High	The authors claim only what they tested, admit a small effect is possible and warn against applying the result to other insects.

Answer key

Study C: Instagram use raises anxiety in teenagers, survey finds

Expert rating: 4 out of 7 · Medium quality

A large survey in a peer-reviewed journal gives a fair base. But teenagers reported their own Instagram use and anxiety only once, which can show a link but not a cause. The title and authors lean towards cause, the journal seeks attention and the paper needs a login. Every criterion lands in the middle, so the study does too.

Criterion by criterion

Criterion	Band	Note
Title	Medium	'Survey finds' is honest about the design, but 'raises anxiety' claims a cause that a one-off survey cannot show.
Access	Medium	Only some people can read it, because the full paper needs a university library login.
Source	Medium	It is peer reviewed, but the journal says it wants 'findings that get people talking' and sends press releases about most papers.
Theory	Medium	It uses the known idea of social comparison but assumes Instagram leads straight to anxiety and ignores other causes of teenage anxiety.
Methods & Data	Medium	Surveying 5,000 teenagers gives a large sample, but each was asked only once about their own hours and anxiety, with raw data only on request.
Conclusion	Medium	Calling the result a link fits the data, but the authors hint at cause and never ask whether anxious teenagers simply use social media more.

Answer key

Study D: Male bosses of the 500 largest US companies are taller than the average US man

Expert rating: 2 out of 7 · Low quality

An exact title and free access are real strengths. But the paper has not been peer reviewed, it offers no evidence for its evolutionary story about the height gap, it ignores the halo effect (height colouring how we judge people) and it claims tall men are born to lead. These faults weigh far more than its strengths.

Criterion by criterion

Criterion	Band	Note
Title	High	It states only what was measured, that male bosses of the 500 largest US companies are taller than the average US man, with no claim about why.
Access	High	Anyone can read the preprint free online, though being free to read says nothing about whether its claims are correct.
Source	Medium	It is a preprint on a public server, so no independent experts have checked it (peer review) and no journal has accepted it yet.
Theory	Low	It says a hunter-gatherer instinct picks bosses but gives no evidence and ignores the halo effect (one liked trait, such as height, colouring how we judge someone).
Methods & Data	Medium	Comparing the heights of 452 bosses with the national average can show a gap, but nothing measured leadership skill or profits and the data are only on request.
Conclusion	Low	Claiming that tall men are born to lead because of evolution treats a height gap as proof of inborn ability, when people may simply tend to pick tall men.

Answer key

Study E: Systematic review of 38 trials: mindfulness brings small to moderate chronic pain relief

Expert rating: 7 out of 7 · High quality

It is free to read and comes from a peer-reviewed library of health reviews. It combines 38 trials with 3,540 people under a preregistered plan, with two reviewers checking each trial for bias. All data and code are open and the conclusion reports how widely results varied. Strong on every criterion, it sits at the top of the scale.

Criterion by criterion

Criterion	Band	Note
Title	High	It names the design, the 38 trials and the condition and says the relief was small to moderate, with no promise of a cure.
Access	High	Anyone can read the full review and download its data free online, so readers can check exactly how it was done.
Source	High	It appeared in a library of health reviews run by an international non-profit network, where each review's plan and final text are peer reviewed.
Theory	High	It brings together many randomised controlled trials with mixed results to answer one clear question, drawing on research into how the brain handles pain.
Methods & Data	High	A preregistered meta-analysis of 38 trials with 3,540 people, separate bias checks by two reviewers and all extracted data and code free online make this careful, open work.
Conclusion	High	It reports a small to moderate drop in pain that varied widely, notes some kinds of pain improved more and says mindfulness can support medical treatment but not replace it.

Answer key

Study F: Real-world study of 5,000 people finds PrEP stops HIV infection

Expert rating: 6 out of 7 · High quality

Free access, a peer-reviewed society journal, a real-world test of earlier trials and blood tests confirming pill use in 5,000 people stand out. It falls just short of the top because users were compared with past patients and only an extra file admits the groups may have differed. Data come only on request and the title leaves out daily use.

Criterion by criterion

Criterion	Band	Note
Title	Medium	It names the real-world setting and sample size without hype words, but 'stops HIV infection' leaves out that protection depended on daily use and was judged against past patients.
Access	High	Anyone can read the full paper free online, so every reader can check the methods and claims for themselves.
Source	High	It appeared in a peer-reviewed infectious disease journal run by a large medical society, with funding from a government health agency.
Theory	High	Earlier randomised trials showed PrEP works when taken as prescribed, so testing how well it works in everyday life is a needed next step.
Methods & Data	Medium	Checking pill use in 5,000 people with blood tests is a strength, but data are only on request and the comparison is past patients, not a group chosen by chance.
Conclusion	Medium	It admits protection falls with missed doses and PrEP does not stop other sexually transmitted infections, but mentions only in an extra online file that past patients may have differed.

Made from the studies, glossary, rubric and rating guide of content version 3 of Unlock the Lab, the same files the app uses. CC BY 4.0, Pablo Bernabeu. https://unlock-the-lab.web.app